All Upper-Critical Trace Coefficients for the Radial Quadratic Pencil

Proof packet by `agent_lane_06`

22 July 2026

Status. Candidate substantial partial result, likely valid, subject to human review. The theorem is all-order in one of the two critical dimension families, but is restricted to $P(x) = |x|^2$.

1 Source problem

Aboud and Robert study the nonlinear pencil

$$L(\lambda) = -\Delta + (P(x) - \lambda)^2$$

for a positive elliptic polynomial P on $\mathbb{R}^d$. Their semiclassical trace formula has coefficients $C_{2j}^{(d)}(f)$. In odd dimensions they prove $C_{2j}^{(d)}(f) = 0$ when $d \geq 4j + 1$, and conjecture that the next two coefficients do not vanish: for every $j \geq 1$, there should be an admissible f for which

$$C_{2j}^{(4j-1)}(f) \neq 0, \qquad C_{2j}^{(4j-3)}(f) \neq 0.$$

This is Conjecture (4.25) of [1].

Now, assume that $2j + 2 \leq k \leq 3j$. Using Lemma 3.2, we have

$$Q_k^{2j}(x, P(x) - z, \xi) = \sum_{\nu + |\gamma| \geq 2(2k-1-2j)} R_{\nu,\gamma}(x)(P(x) - z)^{\nu} \xi^{\gamma}.$$

As above, we integrate by parts in z to have the possibility to put ν at 0 and then we use ξ^{γ} to decrease the order of the singularity in ξ as far as possible (integrability near $\xi = 0$) of $\oint_{\Gamma} \frac{Q_k^{2j}}{L^{k+1}} dz$. We conclude by the analytic dilation argument. $\square$

So, we have proven that in odd dimension d, $C_{2j}^{(d)}(f) = 0$ if $2j \leq \frac{d-1}{2}$.
We conjecture that the next following terms are not 0; more precisely we claim:
Conjecture: For every $j \in \mathbb{N}$, $j \geq 1$, there exists f satisfying (4.14) such that we have we have

$$C_{2j}^{(4j-1)}(f) \neq 0, \quad \text{and} \quad C_{2j}^{(4j-3)}(f) \neq 0 \tag{4.25}$$

Figure 1: Conjecture (4.25) in the source, PDF page 8.

2 Result

Put

$$s_j = [t^{2j}]\frac{t}{\sinh t} = \frac{2(1-2^{2j-1})B_{2j}}{(2j)!}.$$

Theorem 1. *Let $j \geq 1$, $d = 4j-1$, and $P(x) = |x|^2$. For every source-admissible test function f for which the displayed moment converges,*

$$C_{2j}^{(4j-1)}(f) = A_j \int_0^\infty a^{3j-5/2} f(a)\,\mathrm{d}a, \tag{1}$$

where

$$A_j = (-1)^j \frac{6(2j-1)\sqrt{\pi}}{2^{2j+1}\Gamma(2j-\frac{1}{2})} s_j > 0. \tag{2}$$

Consequently $C_{2j}^{(4j-1)}((z+1)^{-(6j-1)}) > 0$ for every $j \geq 1$. The pencil

$$-\Delta + (|x|^2 - \lambda)^2$$

therefore has infinitely many nonlinear eigenvalues in every dimension $d = 4j-1$. For every $\varepsilon > 0$, its counting function satisfies

$$N(R) \geq c_\varepsilon R^{3j-3/2-\varepsilon}$$

for all sufficiently large R.

The sign assertion follows because $\operatorname{sgn} B_{2j} = (-1)^{j+1}$ and $1 - 2^{2j-1} < 0$, so $(-1)^j s_j > 0$. Notice that $6j - 1 > 3(4j-1)/2$, exactly the source's decay condition for the quadratic case.

3 Proof intuition

Rotational symmetry reduces the Weyl-parametrix symbols to four invariants: $a = |x|^2$, $b = |\xi|^2$, $c = x \cdot \xi$, and $w = a - z$. The recurrence closes at every order because the scalar symbol has no derivatives above order four.

The decisive observation is more rigid. Align the first coordinate with x and write $u = \xi \cdot x/|x|$. In the two highest powers of a, one recurrence step is simply

$$a(\partial_w^2 + \partial_u^2).$$

Thus the all-order problem becomes a two-dimensional harmonic-oscillator resolvent in (w, u); the other $d-1$ components of ξ are passive Gaussian variables. Mehler's formula evaluates its generating function. A change of variables $y = \tanh t$ turns the relevant coefficient into a residue of $t/\sinh t$. The Bernoulli numbers then prove nonvanishing at every order, not merely at the finitely many orders accessible by brute-force expansion.

4 Invariant recurrence

Set

$$a = |x|^2, \quad b = |\xi|^2, \quad c = x \cdot \xi, \quad w = a - z, \quad \ell = b + w^2.$$

For an invariant $F(a, b, c, w)$, let $D_a = \partial_a + \partial_w$. Direct differentiation gives

$$\Delta_x F = 2dD_a F + 4aD_a^2 F + 4cD_a\partial_c F + b\partial_c^2 F,$$
$$\Delta_\xi F = 2d\partial_b F + 4b\partial_b^2 F + 4c\partial_b\partial_c F + a\partial_c^2 F.$$

Define

$$S_2 F = \frac{1}{4}\Delta_x F + \frac{w}{2}\Delta_\xi F + 2aF_b + 4c^2 F_{bb} + 4acF_{bc} + a^2 F_{cc},$$
$$S_4 F = \frac{1}{16}\Delta_\xi^2 F.$$

The source's Weyl recurrence specializes, for every $n \geq 1$, to

$$K_0 = \ell^{-1}, \qquad K_{2n} = -K_0\big(S_2 K_{2n-2} + S_4 K_{2n-4}\big), \qquad K_{-2} = 0. \tag{3}$$

Indeed, the fourth x derivative of ℓ yields S_4, while all higher derivatives and all mixed x, ξ derivatives vanish. As a convention audit, (3) gives

$$K_2 = \frac{2dw + 4a}{\ell^3} + \frac{-4wb - 8c^2 - 8aw^2}{\ell^4},$$

which is the source's displayed K_2 formula for $P = |x|^2$.

5 The top-two residue projection

We use the source's normalized contour. Spherical averaging gives

$$\mathrm{av}(c^{2k}) = a^k b^k \frac{(1/2)_k}{(d/2)_k},$$

and the subsequent radial integral is

$$\int_{\mathbb{R}^d} \frac{b^r}{(b+w^2)^q}\, \mathrm{d}\xi = \frac{\pi^{d/2}\Gamma(r+d/2)\Gamma(q-r-d/2)}{\Gamma(d/2)\Gamma(q)}(w^2)^{r+d/2-q}. \tag{4}$$

Lemma 1 (top-two projection)**.** *Let $d = 4j-1$. There is a scalar H_j such that*

$$\int_{\mathbb{R}^d} \oint wK_{2j}(x,\xi,z)f(z)\,\mathrm{d}z\,\mathrm{d}\xi = -H_j\left(a^{j-1}f(a) + \frac{a^j}{3j}f'(a)\right), \tag{5}$$

and

$$H_j = \pi^{2j}(-1)^j 6\, 4^{j-1} j(2j-1)s_j. \tag{6}$$

Proof. First keep only the two highest powers of a in (3). Put $R = |x|$, align $e_1 = x/R$, write $u = \xi \cdot e_1$, and decompose $\xi = ue_1 + v$. At fixed z and ξ,

$$\begin{aligned}\Delta_x = (\partial_R + 2R\partial_w)^2 &+ \frac{d-1}{R}(\partial_R + 2R\partial_w)\\ &+ \frac{1}{R^2}((|\xi|^2 - u^2)\partial_u^2 - (d-1)u\partial_u),\end{aligned}$$

while $(x\cdot\partial_\xi)^2 = R^2\partial_u^2$. Hence, if the top part of K_{2n} is $R^{2n}g_n(w,u,v)$, then

$$g_0 = \ell^{-1}, \qquad g_n = -\ell^{-1}(\partial_w^2 + \partial_u^2)g_{n-1}. \tag{7}$$

The S_4 term and the spherical part of Δ_x are two powers of R lower and do not enter this equation. Keeping one further power gives, for $K_{2n} = R^{2n}g_n + R^{2n-2}r_n + O(R^{2n-4})$,

$$\begin{aligned}r_n = -\ell^{-1}\{&(\partial_w^2 + \partial_u^2)r_{n-1}\\ &+ [\,(2n-2+d/2)\partial_w + (w/2)\Delta_\xi\,]g_{n-1}\}.\end{aligned} \tag{8}$$

Equations (7)–(8), followed by the beta shift (4), are an induction on n. At $n = j$, terms other than $a^j w^{-2}$ and $a^{j-1}w^{-1}$ have zero normalized contour integral. Matching these two terms in (8), and integrating the Δ_ξ term by parts, gives

$$[a^{j-1}w^{-1}] = -3j[a^j w^{-2}]. \tag{9}$$

Because $\oint w^{-1}f(a-w)\,\mathrm{d}z = -f(a)$ and $\oint w^{-2}f(a-w)\,\mathrm{d}z = f'(a)$, this is exactly the ratio in (5). This coefficient comparison uses only $\Gamma(x+1) = x\Gamma(x)$; the packet's exact checker performs it in the unprojected four-invariant recurrence through $j = 7$.

It remains to evaluate the $a^j w^{-2}$ coefficient. Introduce a formal h in (7). The series $G_h = \sum_{n\geq 0}(-1)^n h^{2n} g_n$ solves

$$(-h^2(\partial_w^2 + \partial_u^2) + w^2 + u^2 + |v|^2)G_h = 1.$$

The elementary two-dimensional Mehler kernel, integrated against the constant function, yields

$$G_h = \int_0^\infty e^{-t|v|^2} \frac{1}{\cosh(2ht)} \exp\left[-\frac{\tanh(2ht)}{2h}(w^2+u^2)\right] \mathrm{d}t.$$

After integrating in (u, v) and changing from t to $s = w^2 \tanh(2ht)/(2h)$, the multiplier of the Mellin integral is

$$\left(\frac{y}{\operatorname{artanh} y}\right)^{(d-1)/2} (1-y^2)^{-1/2}, \qquad y = \frac{2hs}{w^2}.$$

For $d = 4j-1$, the coefficient needed at order h^{2j} is therefore

$$F_j = [y^{2j}] \left(\frac{y}{\operatorname{artanh} y}\right)^{2j-1} (1-y^2)^{-1/2}.$$

This coefficient is explicit. In its residue formula put $y = \tanh t$:

$$\begin{aligned} F_j &= \operatorname*{Res}_{y=0} y^{-2j-1} \left(\frac{y}{\operatorname{artanh} y}\right)^{2j-1} (1-y^2)^{-1/2}\,\mathrm{d}y \\ &= \operatorname*{Res}_{t=0} t^{-(2j-1)} \frac{\cosh t}{\sinh^2 t}\,\mathrm{d}t \\ &= -(2j-1) \operatorname*{Res}_{t=0} t^{-2j} \frac{1}{\sinh t}\,\mathrm{d}t = -(2j-1)s_j. \end{aligned}$$

Here the third equality is integration by parts in the formal residue. Restoring the Gaussian factor $\pi^{d/2} 4^j \Gamma(3/2)$ and the sign between G_h and g_j gives the $a^j w^{-2}$ coefficient

$$-\frac{H_j}{3j} = (-1)^{j+1} \pi^{2j} 2^{2j-1} (2j-1) s_j.$$

This is equivalent to (6), completing the induction and the proof. □

6 Completion of the trace computation

The source's trace prefactor is $-2(2\pi)^{-d}$. In polar coordinates,

$$\int_{\mathbb{R}^d} \Phi(|x|^2)\,\mathrm{d}x = \frac{\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty a^{d/2-1} \Phi(a)\,\mathrm{d}a.$$

Insert Lemma 1. Integration by parts gives

$$\begin{aligned} &\int_0^\infty a^{d/2-1} \left(a^{j-1} f(a) + \frac{a^j}{3j} f'(a)\right) \mathrm{d}a \\ &\quad = \left(1 - \frac{3j-3/2}{3j}\right) \int_0^\infty a^{3j-5/2} f(a)\,\mathrm{d}a = \frac{1}{2j} \int_0^\infty a^{3j-5/2} f(a)\,\mathrm{d}a. \end{aligned}$$

Consequently

$$C_{2j}^{(4j-1)}(f) = \frac{H_j}{j} (2\pi)^{-d} \frac{\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty a^{3j-5/2} f(a)\,\mathrm{d}a.$$

Substituting $d = 4j-1$ and (6) simplifies the prefactor to (2). This proves the trace formula and its positivity. The nonlinear-eigenvalue and counting conclusions are then exactly the source's trace criterion and scaling corollary, with exponent

$$(d-2j)\frac{2+1}{2} = \frac{3}{2}(2j-1) = 3j - \frac{3}{2}.$$

This proves Theorem 1. □

7 Exact verification

The script `code/verify_all_order_top.py` reconstructs the complete four-invariant recurrence, performs the angular averages and beta integrals, and checks (6) and (9) exactly for $1 \le j \le 7$. At $j = 7$ the unprojected numerator has 238 monomials. The command

```
conda run --no-capture-output -n sandbox python \
  code/verify_all_order_top.py --through 7
```

ends successfully; the recorded output is in `verification.md`. This is an audit of the arbitrary-j coefficient proof, not a finite-order substitute for it.

8 Scope and novelty check

The theorem settles the full upper-critical family $d = 4j - 1$ for the radial quadratic polynomial. It does not settle the lower family $d = 4j - 3$ or the conjecture for arbitrary positive elliptic P. Exact experiments find the lower-family coefficient positive through $j = 6$, but no all-order sign identity was found, so that pattern is not claimed as a theorem.

A bounded novelty search on 22 July 2026 checked the run's four cheap indexes and arXiv searches for the source title and authors together with the phrases "$4j - 1$ coefficient", "radial quadratic", "Bernoulli", and "nonlinear eigenvalue pencil". It found the source, its authors' later numerical work, and general nonlinear-eigenvalue reviews, but no analytic all-order computation of this coefficient family. Novelty confidence is moderate, not definitive, because notation-specific searches can miss an equivalent result.

9 Human-review recommendation

Review Lemma 1 first. In particular, check the top-two R-power expansion (7)–(8), the ratio (9), and the contour signs. Next check the Mehler normalization and the $y = \tanh t$ residue. Finally rerun the exact checker; its agreement through seven complete, rapidly growing symbols is a strong convention and algebra audit.

References

[1] F. Aboud and D. Robert, *Asymptotic expansion for nonlinear eigenvalue problems*, J. Math. Pures Appl. 93 (2010), 149–162, arXiv:0903.0919, https://arxiv.org/abs/0903.0919.